PEQG26
Population, Evolutionary and Quantitative Genetics Conference

ID: 474

How Cancer Cells Undergoing a Treatment Course Evolve to Survive Multiple Mass Extinctions

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ABSTRACT

The vast diversity of genetic and phenotypic characteristics observed in human cancers, according to Hanahan (2026), can be understood in the context of how cancers arise via multistep pathways of tumorigenesis that lead to the formation of primary tumors, which subsequently evolve and progress due to selective advantage in the face of multifaceted barriers to continuing proliferative expansion, resulting in metastasis and adaptive resistance.

Here we elucidate the adversarial viewpoint: a *press-pulse* framework borrowed from paleobiology (Arens & West, 2008) for modeling therapeutic intervention targeted at the primary tumor, whose microenvironment undergoes *terraforming* and subsequently becomes unfit for cancer cells proliferation, leading to multiple *mass extinctions* over the course of cancer treatment, finally resulting in remission as the residual cancer cells turn quiescent.

The population dynamics driven by tumor growth vs. dose response is captured in our proposed model, as the cancer cells population follows a Gompertzian growth curve after metabolic reprogramming but shrinks in response to drug treatment. To shrink a tumor over multiple treatment cycles, dose responses are modeled as mass extinctions of cancer cells in our testbed as intravenous antibody-drug conjugate (Madhusoodanan, 2024) diffuses across blood vessel walls to varying tumor depths, driven by concentration gradient according to Fick's laws.

Tumoral evolution within our framework offers mechanistic insights into the structure of a shifting fitness landscape characterized by dynamic dose responses and residual regrowth. We illustrate how residual subpopulations of cancer cells evolve to survive multiple mass extinctions originated from a prescribed course of antibody-drug conjugate targeting the primary tumor, potentially laying the seeds for future metastasis or adaptive resistance in the unfortunate event of a relapse. We propose to construct a simulation testbed based upon this key insight from population dynamics for further exploration.

INTRODUCTION

Antibody-Drug Conjugates Show Superior Response when Dosed at MTD

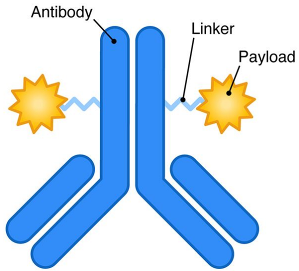


Fig. 1: Antibody-Drug Conjugate (ADC) [4]

Over the last two decades, an increasing number of antibody-drug conjugates (ADCs) have been approved for cancer treatment, broadening therapeutic options for patients beyond traditional small molecule drugs. Through their unique mode of action as *“magic bullets”*, these targeted therapies hold the promise of redressing the poor balance between *safety* and *efficacy* with traditional chemotherapy [4, 6].

Image Credit: Drago & Norton (2021)

The "sweet spot" between the lowest concentration that produces a desired therapeutic effect (minimum effective dose or MED) and the concentration just before toxicity kicks in (maximum tolerated dose or MTD) is called the *therapeutic window*.

When dosed at or near MTD, ADCs display improved efficacy over small molecules in trials. Across the board, ADCs have already achieved greater beneficial outcomes for patients compared with traditional chemotherapy. The *objective response rates* (ORR) of ADC treatments frequently score above 40 percent in breast, gastric, urothelial, and lung cancer trials where small molecules rarely exceed 25 percent [2, 8].

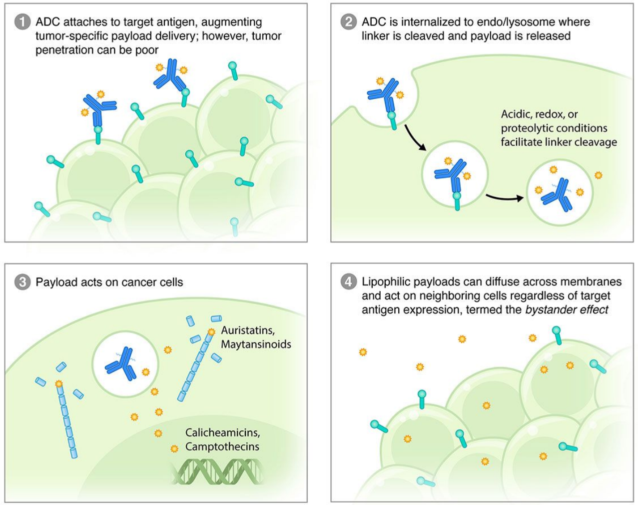


Fig. 2: Mechanism of Action for Antibody-Drug Conjugates [4].

RESEARCH QUESTION

Terraforming a Tumor Microenvironment for "Worst Fit" Under Press-Pulse?

We've identified three interacting forces that drive cancer cells population dynamics during a treatment course:

(a) **Growth**: proliferative cancer cells utilize aerobic glycolysis to generate energy from glucose at a fast rate to drive a Gompertzian tumor growth curve;

(b) **Treatment**: dose responses are modeled as mass extinctions of cancer cells as intravenous drug diffuses across blood vessel walls to varying tumor depths, driven by concentration gradient according to Fick's laws; and

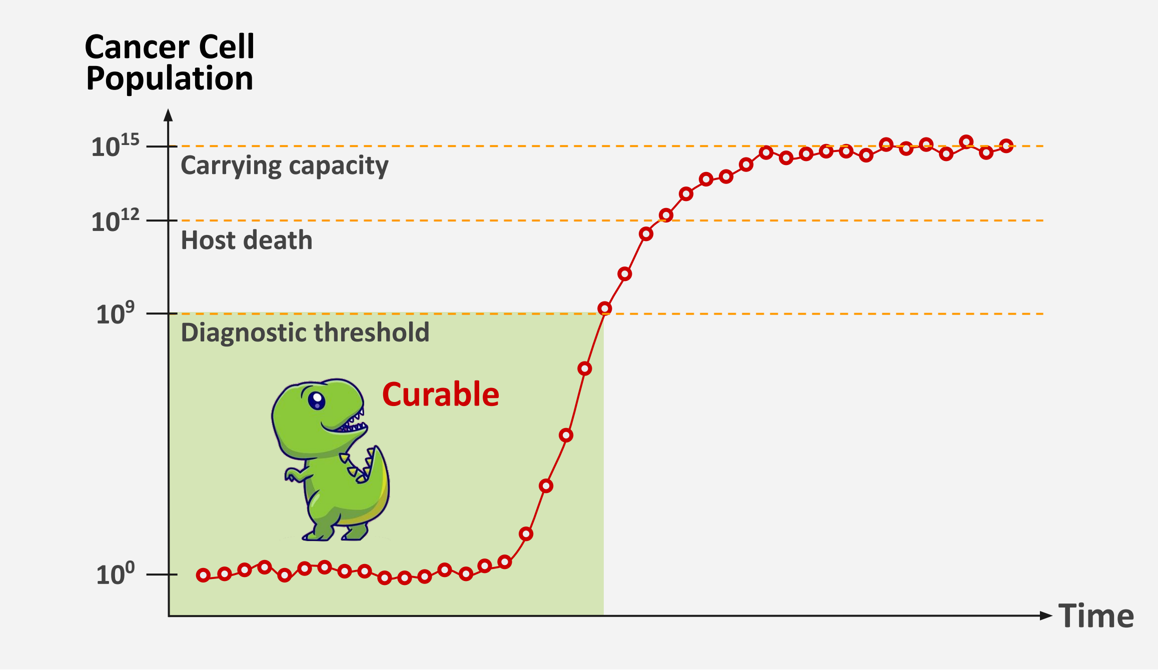
(c) **Mutation:** somatic mutation during residual regrowth in between treatment cycles selects for drug-resistant cells that drive future cancer recurrence.

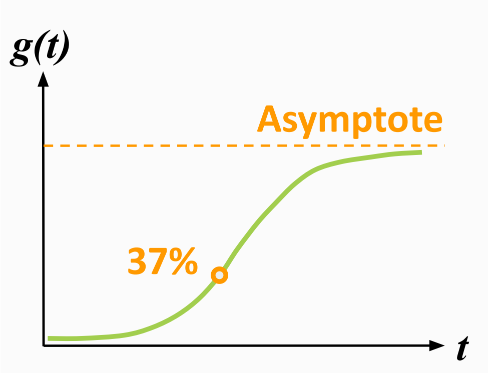
Fig. 3: **Terraforming** a tumor microenvironment under press-pulse.

Key Insight: We have always been fascinated by the idea of tumoral evolution. Cancer cells compete for resources in order to grow rapidly, while evading counterattacks. Mathematical models describe how cancer cell populations grow with aerobic glycolysis but shrink under drug treatment, just as Lotka-Volterra equations describe the population dynamics of snowshoe hares and the lynx that feed upon them. We introduce new complexity into population dynamics by adding somatic mutation across treatment cycles.

METHODS — Growth

Modeling Tumor Growth with a Gompertzian Curve





$$g(t) = a \cdot e^{b \cdot e^{ct}}$$

Fig. 4: **Gompertzian tumor growth:** initial growth at the inflection point appeared exponential, before slowing to approach the asymptote (i.e., asymmetric sigmoid curve).

METHODS — Treatment

1. Dose at Maximal Effective Concentration — Dependent on Size of Tumor

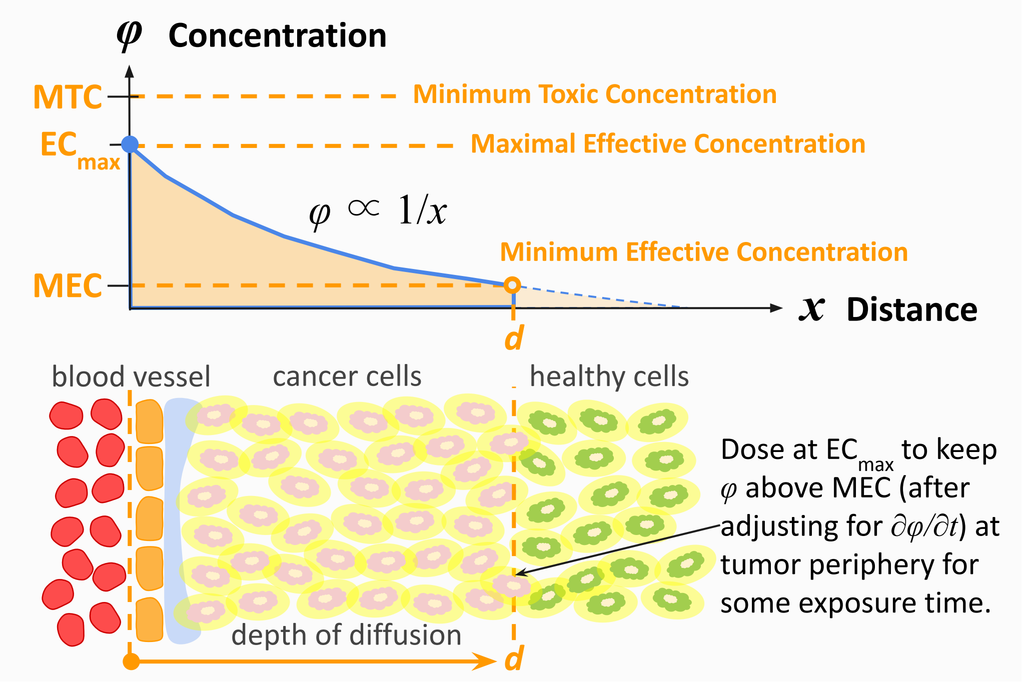


Fig. 5: **Fick's laws** describe diffusion driven by a concentration gradient, from high-concentration intravenous space to low-concentration intercellular space.

$$J = -D\,\frac{d\varphi}{dx}$$

Fick's 1st law

where

- J is the diffusion flux, which measures the amount of substance flowing through a unit area during a unit time interval,
- D is the diffusion coefficient, or diffusivity, in area per unit time.

$$\frac{d\varphi}{dx} = \frac{1}{x^2}$$

Concentration gradient formula

METHODS — Treatment

2. Compute Dose Response Curve — for Each Tumor Size and Shape

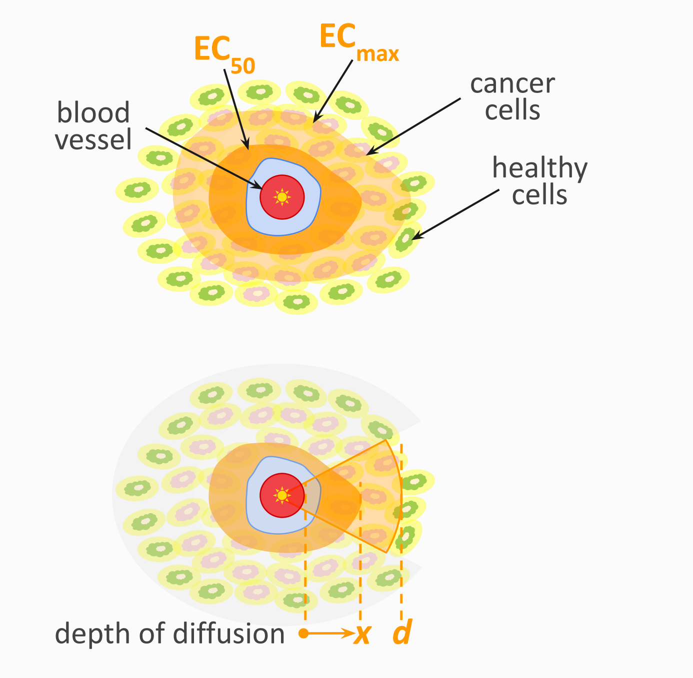


Fig. 6: **Cross-sectional view:** depth of diffusion into a tumor determines effective concentration of dose.

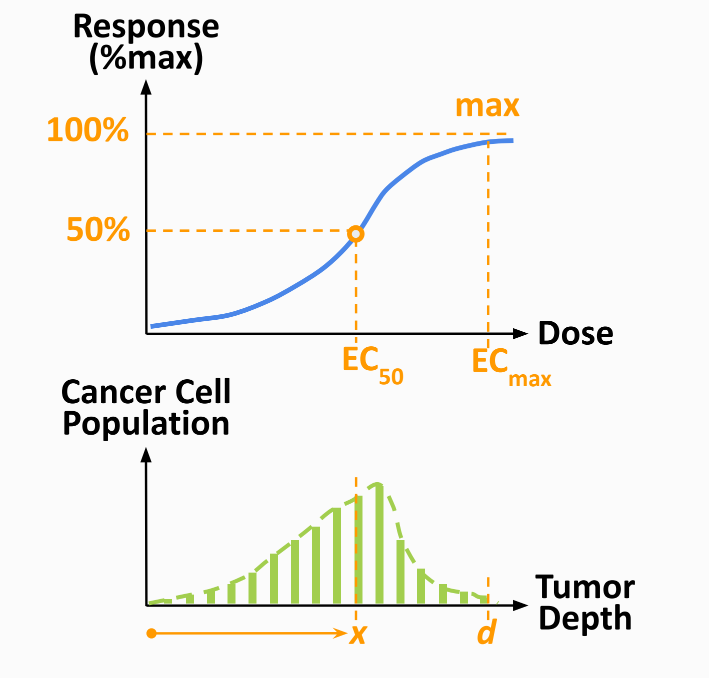


Fig. 7: **Dose response curve** computed based on cancer cell population reachable at each tumor depth.

METHODS — Treatment

3. Adjust Dose Response Model — as Concentration Decreases Over Time

$$Exposure\ Time = t_1 - t_0$$

$$\frac{\partial \varphi}{\partial t} = D\frac{\partial^2 \varphi}{\partial x^2}$$

Fick's 2nd law

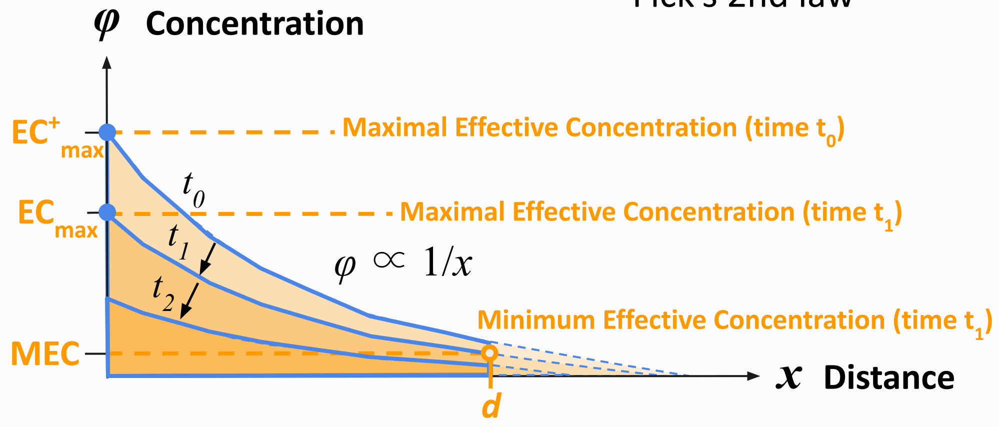


Fig. 8: **Changing concentration:** ensure tumor periphery at *d* has enough exposure time (t_1 - t_0) by working backwards to set maximal effective concentration at EC^{+}_{max}.

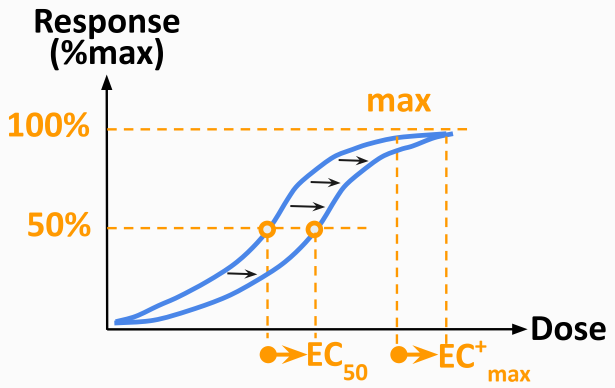


Fig. 9: Dose response model adjusted for maximal effective concentration (at EC^{+}_{max}).

DISCUSSION — Mutation

Cancer Cells Evolve in Between Treatment Cycles to Survive Mass Extinctions

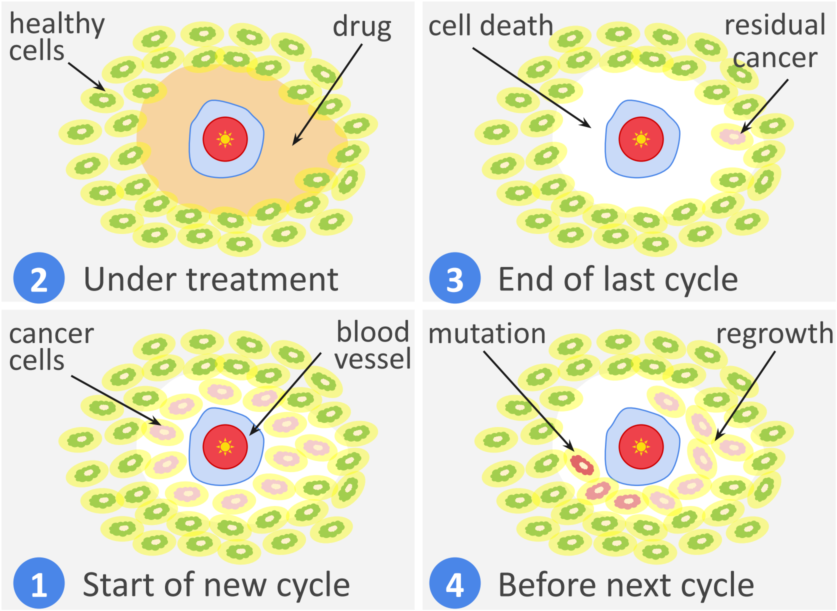


Fig. 10: **Residual regrowth** in between treatment cycles selects for drug-resistant cells that drive future cancer recurrence [17].

Would starving *proliferative* cancer cells of glucose improve dose response for cancer patients?

Could one switch to a *reduced dose* regimen without compromising dose response by going on a calorie-restricted diet to bring down the glucose level, which slows down the cell cycle and makes proliferative cancer cells more vulnerable?

One could then apply a *denser* dose schedule to suppress residual regrowth while shortening treatment cycles.

Furthermore, reduced heterogeneity in the tumor would then make relapse less likely, thus maximizing disease-free survival (DFS).

CONCLUSIONS

That's Why We Should Minimize Somatic Mutation Across Treatment Cycles!

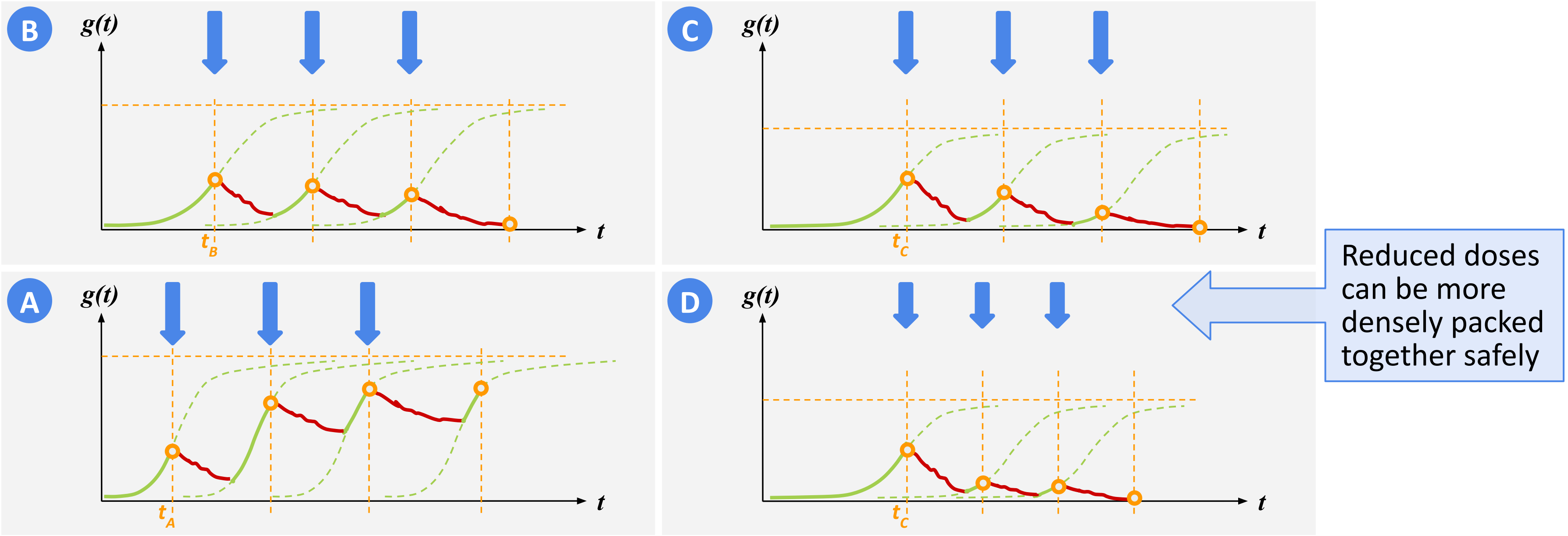


Fig. 11: Path dependency in how tumor size changes across multiple treatment cycles: (A) high residual regrowth, (B) baseline residual regrowth, (C) low residual regrowth, and (D) even lower residual regrowth with a *denser* dose schedule. Somatic mutation is minimized when residual regrowth is suppressed during a course of treatment.

FUTURE WORK

Safer ADC Regimen That Maximizes Disease-Free Survival?

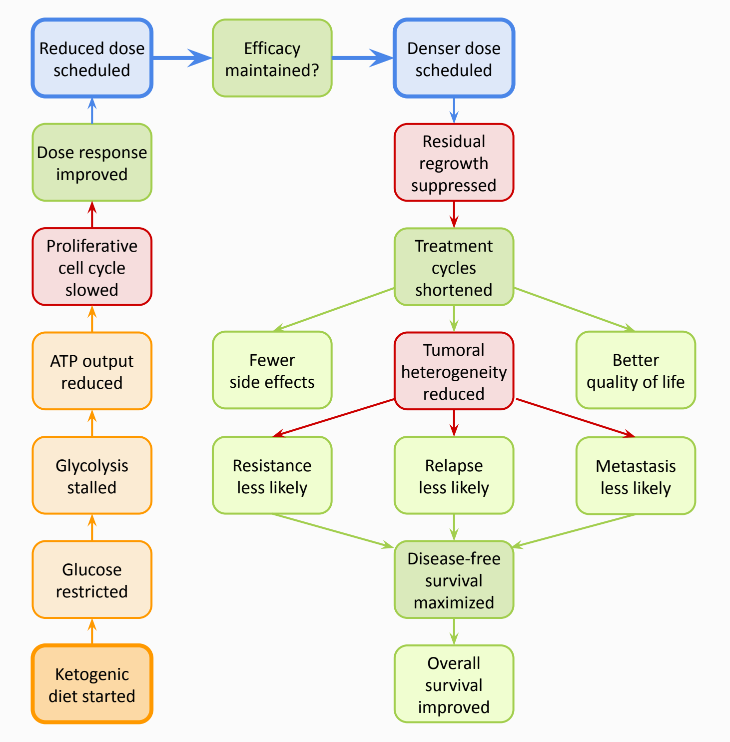


Fig. 12: *Chain-of-thought reasoning* for our research design.

We propose that tumor size and bioenergetics together determine dose response, which we see as multifaceted and not simply a static curve as commonly believed. Importantly, a cancer patient can improve dose response with nutritional intervention under one's control.

Our ongoing research has therapeutic implications: a *safer* ADC regimen realized through improved dose response modeling benefits not just late-stage cancer patients considering palliative care, but also early-stage cancer patients seeking alternative treatment paths that maximize long-term *disease-free survival* [3].

By taking a longer view of DFS as an alternate endpoint, we hope to optimize *safety* and *efficacy* together by limiting residual regrowth during treatment — unlike traditional chemotherapy preoccupied with maximizing cell kill in the primary tumor, sacrificing safety.

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